Forecasting Next-Week Volatility of S&P 500 Stocks

Milestone 2: data wrangling and project redefinition

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CS1090A / AC209A, Fall 2026 Group 38, project 12 TF: Daniel More Torres

The question and why it matters

Ashley

- Returns are close to unpredictable; **volatility clusters**, so it can be forecast.
- Volatility forecasts size positions, set risk limits and price options.
- HAR (Corsi, 2009), a regression on last day, week and month of volatility, is hard to beat.
- **Our question**: with free daily data, does one pooled model with cross-sectional features beat HAR fit stock by stock, out of sample, and in which market regimes?

The data: five public sources (473 MB) and a CRSP check

Source	What	How
fja05680/sp500	S&P 500 members on 2,720 dates since 1996	CSV, pinned
Yahoo Finance	daily OHLCV 2005–2026 for 677 of 950 past and present members	`yfinance`
Alpaca	daily bars from 2016 for 165 members Yahoo lacks; 5-minute bars	REST API
FRED	VIX, 3-month VIX, credit spread, curve slope, T-bill	CSV
SEC EDGAR	earnings-release filing times (8-K Item 2.02), industry codes	JSON API
CRSP (WRDS)	every member's daily prices 2005–2025, for the survivorship check only	SQL, licensed

- Every fetch is cached and re-runnable; no data is committed (licences).
- Code in `src/volforecast`, unit tests incl. no-look-ahead checks, CI on every pull request.

What one row is

Raw table: one ticker on one trading day

3,423,011 rows: `date`, `ticker`, `open`, `high`, `low`, `close`, `adj_close`, `volume`, `source`

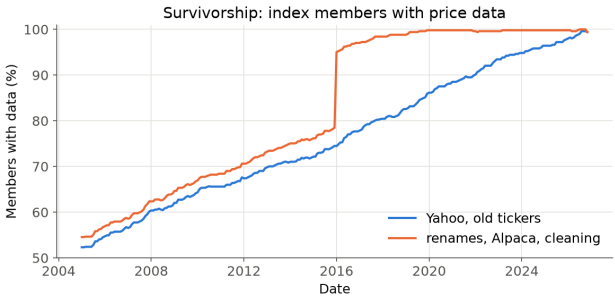
Modeling table: one stock in one week

473,684 rows $\times$ 183 columns, 725 companies, 1,139 weeks

- target: log of next week's Garman–Klass variance
- HAR lags: day, week, 4 weeks
- range, overnight, downside variance; returns; liquidity
- market: SPY variance, VIX, rates
- earnings timing, industry; Alpha158 (158 price-volume factors)

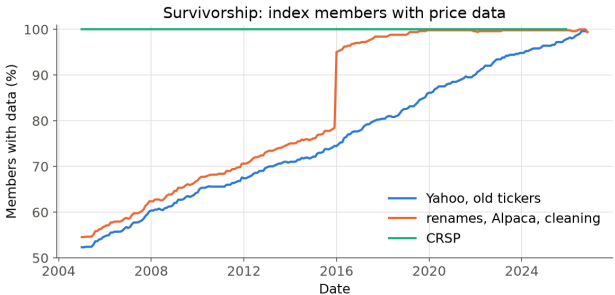
Every feature is known at that Friday's close.

Survivorship: who is missing, and when



- Yahoo drops delisted firms and files renamed ones under the new symbol.
- 30 renames recovered (BK $\rightarrow$ BNY, ANTM $\rightarrow$ ELV, ...).
- Alpaca fills members that traded in 2016 or later.
- Coverage: 55% in 2005, 99.4% in 2019, 99.8% in 2026.
- Pre-2016 training data leans toward survivors.

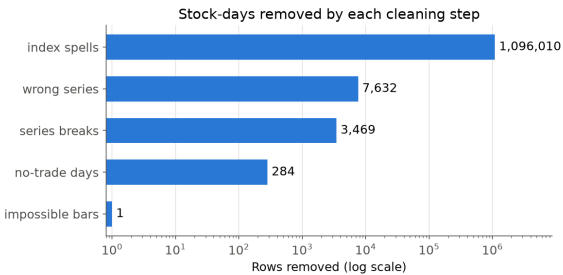
What survivorship costs: a CRSP check



- CRSP through Harvard's WRDS licence: every member, every day, 2005–2025.
- The same cleaning and weekly table, run on CRSP.
- 2005–2015: 191,083 stock-weeks in the free data, 286,597 in CRSP.
- Survivors are calmer: mean log variance 0.04 lower before 2016, within 0.01 after.
- It costs training data more than it biases the target; the test period is untouched.

Cleaning: each rule, and what it removes

Shuo

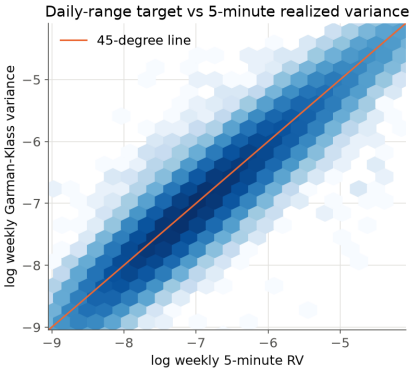


1. **Index spells**: Yahoo reuses tickers (today's `FB` is an ETF).
2. **Wrong series**: 9 hand-checked (Yahoo's PARA trades near $100,000); 4 series that change company (Yahoo's JCI before 2016 is Tyco).
3. **Impossible bars** and **no-trade days**.
4. Prices scaled by `adj_close / close`, as qlib does.

3,423,011 $\rightarrow$ 2,315,615 stock-days. No rule looks at later rows.

The target, checked against 5-minute data

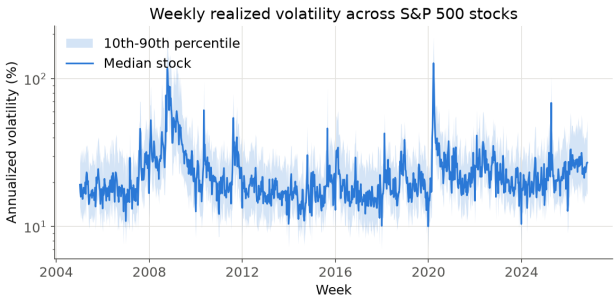
Simon



- Target: $\log \sum_{d \in \text{next week}} \left[\frac{1}{2} \log^2 \frac{H_d}{L_d} - (2 \log 2 - 1) \log^2 \frac{C_d}{O_d} \right]$ (Garman–Klass, 1980).
- The proposal feared daily data is too noisy. Against 5-minute realized variance (Alpaca, 697 companies since 2016, delisted included; 281,250 stock-weeks): correlation of log variances 0.95 pooled, 0.94 within company.
- Log gap -0.06 on average: a level shift that a model absorbs.

Calm and stress: the regression analogue of class imbalance

Simon



- Stress week: VIX ≥ 30 on the Friday, known when we forecast.
- Only 7.3% of stock-weeks (89 weeks), mostly 2008–09, 2020 and 2022.
- We will report results separately for calm and stress weeks.

Information leakage: how each risk is closed

Simon

Risk	Control
windows past the Friday	all windows end at the close; tests truncate data and recompute
earnings after the close	an 8-K filed after 16:00 ET counts for the next week (filing time as proxy)
knowing the next earnings date	extrapolated from past announcement dates only
10-Q dates as earnings dates	not used: filed weeks after the announcement
today's index list	the snapshot in force on each date (the last, 2026-08-18, carried forward)
back-adjusted prices	every price feature is a ratio or a return
overlapping targets	a one-week gap between training and test windows

Revised question

Train 2005–2015, tune 2016–2018, test 2019–2026 with yearly expanding refits.

1. **Accuracy**: do pooled ridge/lasso and gradient-boosted trees beat per-stock HAR on mean QLIKE and MSE? Diebold–Mariano at 5% on the weekly mean loss difference (Newey–West); all models scored on the same rows.

2. **Attribution and regimes**: which feature blocks carry any gain (refit without each block), and does it keep its sign in calm and in stress weeks?

Changed: economic-value backtest dropped; earnings from 8-K timestamps; fundamentals deferred (XBRL starts 2009); Alpha158 added; stress defined by VIX, not calendar years.

Staff notes and the AC209A requirement

Terry

Staff note	What we do
omit economic value, Sharpe, drawdown, factor regression	dropped entirely
prioritize regularized regression and boosted trees	the two pooled models of Q1
clustering and neural networks optional	only if time remains after Q1 and Q2
yfinance can fail; delisted stocks missing	cached fetches; renames, wrong series and Alpaca fill; coverage reported
XBRL only from mid-2009	fundamentals deferred
AC209A: one method not covered in class	HAR, GARCH(1,1), QLIKE loss (Patton, 2011), Diebold–Mariano test

Next steps and questions for you

all

Milestone 3 (proposed owners)

- HAR and GARCH baselines: Simon
- ridge and lasso: Shuo
- gradient-boosted trees: Ashley
- QLIKE, MSE, Diebold–Mariano: Terry
- data upkeep and CRSP: Juntang

Questions

1. May Milestone 3 train on CRSP (WRDS) to 2025 and use free data for 2026?
2. Do HAR, GARCH, QLIKE and Diebold–Mariano meet the AC209A requirement?
3. Yearly expanding refits on 2019–2026, or one fit?